



IILM UNIVERSITY, GREATER NOIDA

School of Computer Science and Engineering

Internship Presentation

Legal/RTI Document Assistant

using Corrective RAG & GraphRAG

STUDENT NAME

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ENROLLMENT NO.

CS-23411301

PROGRAM & SEMESTER

B.Tech CSE (AI/ML & Cybersecurity) – 7th Semester

ORGANIZATION

Vedas Tech, Anand, Gujarat

Introduction & Organization

The Internship

Role: AI/ML Product Engineering Intern at Veda Tech

Duration: 9 June 2026 – 10 August 2026 (2 months)

Built an advanced Retrieval-Augmented Generation (RAG) system — the “Legal/RTI Document Assistant” — an AI-powered tool that gives citizens accurate, citation-grounded answers on RTI Acts, court judgments, and CIC decisions.

Problem Statement

Citizens struggle with complex legal language across RTI Acts and precedents. A system was needed to answer natural-language queries accurately, with proper citations, and without hallucinating information.

ORGANIZATION PROFILE

Veda Tech

Empowering AI & Language Services since 2021

- Headquartered in Anand, Gujarat
- AI-driven products & language-based solutions
- Hands-on, real-world product engineering experience for interns

Objectives & Scope

PROJECT OBJECTIVES

- 1 Index the RTI Act, State RTI Acts, court judgments & CIC decisions
- 2 Retrieve accurate answers with section/clause-level precision
- 3 Generate hallucination-free, citation-grounded responses
- 4 Link related legal precedents through graph-based relationships

SCOPE OF THE PROJECT

Central RTI Act, 2005

Core statute fully indexed

Delhi RTI Act, 2001

State-level rules covered

Delhi High Court judgments

Linked via citation graph

CIC decisions

Precedent-level retrieval

Designed to be extensible to additional state Acts and judgments in future.

Technologies & System Architecture

TECH STACK

● Python

● pdfplumber

● sentence-transformers

● rank-bm25

● Qdrant

● Groq

● python-dotenv

● Git / VS Code

SYSTEM ARCHITECTURE

1

Parent-Child Chunking

Section = parent, clause (e.g. 8(1)(j)) = child

2

Hybrid Retrieval

BM25 keyword search + vector similarity search

3

Self-Grading / Re-retrieval

Corrective loop refines weak retrievals

4

Faithfulness Check

Hallucination check after answer generation

5

Citation Graph Layer

Links Act clauses to related case documents

Methodology

1

Data Collection

RTI Act 2005, Delhi RTI Act 2001, judgments & CIC decisions

2

Extraction & Cleaning

Resolving OCR issues and font-mapping bugs

3

Parent-Child Chunking

Metadata: section, subsection, clause

4

Indexing

Load chunks into the Qdrant vector database

5

Retrieval & Generation

Hybrid BM25 + vector search, answers via Groq

6

Verify & Test

Hallucination checks, citation graph, end-to-end testing

Results & Evaluation

A working Corrective RAG + GraphRAG pipeline delivering accurate, citation-backed answers while automatically detecting and correcting hallucinated information.

EVALUATED USING RAGAS METRICS

Faithfulness

Answer Relevancy

Context Precision

Context Recall

KEY TAKEAWAYS

- Section/clause-level precision reduces manual legal research time
- Citation-grounded, hallucination-checked answers build citizen trust
- Graph layer connects Act clauses to real case precedents
- Modular design is extensible to additional state RTI Acts